



A CT-based radiomics model for classification of chronic pancreatitis: new biomarkers for diagnosis and severity staging

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ABSTRACT

Objectives: To develop and validate CT-based radiomics models for the identification of chronic pancreatitis (CP) and selected CP-related complications, and to explore associations between radiomics-derived features and clinical measures.

Material & methods: This retrospective, multicenter study included a training cohort of 349 subjects (201 CP; 148 healthy controls) from Aalborg University Hospital. Test cohort comprised 109 subjects, including 41 pancreas-healthy controls from Aalborg and an external cohort of 68 CP patients from Bergen. Portal venous phase CT scans were automatically segmented, and radiomics features were extracted using PyRadiomics. AI models were trained to classify CP and identify CP-related complications, including exocrine pancreatic insufficiency (EPI), diabetes, and pain. Model performance was assessed using area under the receiver operating characteristic curve (AUC) with 95% confidence intervals (CI).

Results: The CP classification model demonstrated high discriminative performance with an AUC of 0.97 (95% CI: 0.94–0.99). The EPI model showed moderate discriminative performance (AUC 0.80, 95% CI: 0.66–0.88). In contrast, the diabetes and pain models demonstrated lower discriminative performance, with AUCs of 0.63 (95% CI: 0.47–0.77) and 0.59 (95% CI: 0.37–0.67), respectively. Radiomics-derived probability scores correlated significantly with fecal elastase levels ($p < 0.001$) and increased with greater functional disease severity ($p = 0.004$).

Conclusion: CT-based radiomics can accurately classify CP and reflect exocrine functional impairment. However, performance for diabetes and pain was limited, and clinical utility beyond established clinical assessments remains to be demonstrated.

1. Introduction

Chronic pancreatitis (CP) is an inflammatory disease characterized by progressive and irreversible damage to the pancreas. It imposes a significant burden on patients and healthcare systems due to severe complications, including persistent abdominal pain, diabetes, and exocrine pancreatic insufficiency (EPI) [1,2]. Modern pathophysiology models, especially mechanistic framework suggested by Whitcomb

et al., depict CP as a dynamic continuum that starts with a single episode of acute pancreatitis, evolves through recurrent acute attacks, and with progression to end-stage glandular destruction [3]. However, from an imaging perspective, only the endpoints of this spectrum—acute attacks and late permanent structural damage—can currently be visualized with confidence using cross-sectional imaging modalities such as CT, MRI, and endoscopic ultrasonography. While imaging plays a central role in diagnosing CP, associated complications are typically identified using

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clinical evaluation and laboratory testing. The diagnostic gap in early and intermediate disease stages underscores the need for non-invasive biomarkers that can better stratify patients across the full spectrum of CP [4]. This would allow clinicians to intervene before the development of permanent gland damage and individualize monitoring and therapeutic timing.

Radiomics offers a promising approach to address these challenges. By identifying subtle voxel-level patterns, radiomics may uncover microstructural changes that the human eye and conventional analyses cannot detect. When combined with artificial intelligence (AI), including automatic pancreas segmentation, radiomics can be used to generate classification/predictive models, potentially serving as non-invasive “virtual biopsies” [5]. Although AI and radiomics have been increasingly applied to pancreatic imaging, their use in chronic pancreatitis remains limited, with only few studies exploring CT-based radiomics in CP. Consequently, the potential of quantitative imaging features to reflect CP-related structural changes and their associations with clinical complications such as EPI, diabetes, and pain remains insufficiently explored.

The goal of this study was therefore to evaluate the utility of radiomics as clinically relevant imaging biomarkers in CP. Overall, we aimed to determine whether radiomics features derived from pancreatic computer tomography (CT) in combination with AI based analysis could (1) identify patients with CP compared to healthy subjects and further classify patients according to EPI, diabetes or pain complications, and (2) provide insight into the underlying pathoanatomy associated with these complications. In addition, we (3) explored whether the AI-generated probability scores could be used as a measurement for disease severity, specifically for exocrine insufficiency and functional disease severity.

2. Material & methods

2.1. Ethical approvals

This retrospective, cross-sectional, multi-center study used the CLEAR checklist for reporting (Supplementary 1) [6]. It was approved by the Danish National Committee on Health Research Ethics (Ref: 2308560) and The Regional Ethical Committee in Western Norway (Ref: 2019/1037), with data-sharing agreements between Aalborg University Hospital (Aalborg, Denmark) and Haukeland University Hospital (Bergen, Norway) approved by the Scandinavian Baltic Pancreas Club (SBPC) steering committee.

2.2. Study population

The study included a CP cohort and a pancreas-healthy control cohort from Aalborg as well as an external CP cohort from Bergen. All cohorts were selected using identical inclusion and exclusion criteria, as detailed below. Fig. 1 provides an overview of the study workflow.

2.2.1. Aalborg CP cohort

The training dataset consisted of CT scans from patients diagnosed with CP between January 2012 and December 2022 at Aalborg University Hospital, identified through the SBPC database [7]. The CP diagnosis was based on the M-ANNHEIM (Multiple risk factors, Alcohol consumption, Nicotine consumption, Nutrition, Hereditary factors, Efferent duct factors, Immunological factors, and Miscellaneous and rare genetic factors) criteria [8]. Clinical variables, including the presence of EPI, diabetes and abdominal pain, were extracted from the database within 12 months before or after the CT scan. Diabetes diagnosis was given based on the American Diabetes Association guidelines [9]. EPI diagnosis relied on fecal elastase concentration < 200 µg/g or documented clinical symptoms in combination with response to enzyme replacement therapy [10]. Painful CP was defined as weekly or more frequent abdominal pain based on standardized questionnaires [7]. If

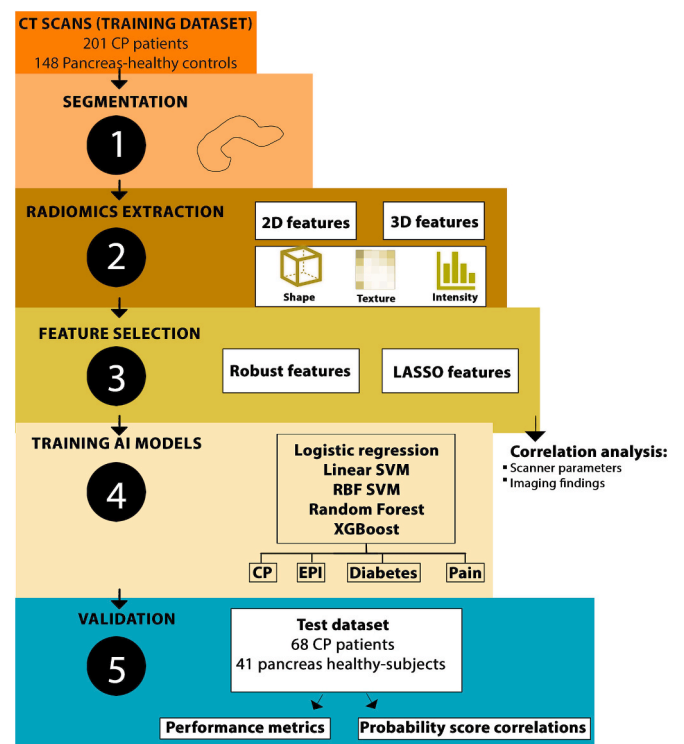


Fig. 1. Overview of the study design and AI model development workflow. The process included: (1) pancreatic segmentation, (2) extraction of radiomic features (shape, intensity, and texture), (3) feature selection with LASSO regression, (4) AI model training, and (5) validation with test dataset.

questionnaire data were unavailable, medical records and opioid prescriptions were reviewed as proxy indicators for pain.

2.2.2. Aalborg control cohort

Pancreas-healthy controls were retrospectively identified from trauma patients who underwent chest-abdominal CT at Aalborg University Hospital between January 2014 and December 2016. Most were allocated to the training cohort, while a smaller portion was withheld for independent testing. To ensure absence of CP, medical records were cross-checked and CT scan was reviewed for pathological conditions that could impact pancreatic anatomy.

2.2.3. Bergen external CP cohort

An external test cohort of CT scans from CP patients, acquired between May 2009 and November 2021, was obtained from Haukeland University Hospital through the SBPC database, meeting identical diagnostic and exclusion criteria as the Aalborg cohort.

Eligibility criteria: Exclusion criteria were identical for all cohorts: 1) Presence of other pancreatic pathologies (e.g., acute pancreatitis, malignancy, pseudocysts > 4 cm), 2) intra-abdominal pathologies (e.g., ascites, malignancy, trauma), 3) CT scans with artifacts obscuring the pancreas, 4) prior pancreatic surgery, 5) presence of pancreatic duct stents, and 6) severe fat infiltration of the pancreas.

2.3. Image acquisition

All CT scans were acquired during the portal venous phase, however, acquisition protocols were not standardized across centers. Scans were obtained using scanners from multiple vendors (including Siemens, GE, and Philips), with slice thickness ranging from approximately 1–5 mm and tube voltages typically between 100 and 140 kVp. Scan coverage varied according to clinical indication (e.g., CT abdomen, CT pancreas protocol, or CT thorax–abdomen–pelvis), and contrast administration

parameters (contrast agent and volume) also differed across institutions. Detailed scanner models and acquisition parameters for each cohort are provided in [Supplementary 2 Section 1](#).

2.4. Automatic segmentation

Automatic pancreatic segmentation of the CT scans was performed using our deep learning tool based on the nnUnet architecture trained on CP patients and healthy controls [11]. All segmentations were reviewed, and adjusted if necessary, by a junior radiologist with 3 years of radiology experience and supervised by a radiologist specialized in pancreas imaging with more than 20 years of experience.

2.5. Radiomics features

A total of 107 radiomic features were extracted using PyRadiomics (version 3.1.0) in Python (Python Software Foundation, version 3.13.2, <https://www.python.org/>), adhering to the International Biomarker Standardization Initiative guidelines [12]. The extracted features included shape-based features describing pancreatic morphology, first-order statistical features quantifying voxel intensity distributions, and texture features characterizing relationships between voxels. Texture features were derived as Gray Level Co-occurrence Matrix, Gray Level Run Length Matrix, Gray Level Size Zone Matrix, Gray Level Dependence Matrix, and Neighbouring Gray Tone Difference Matrix. Higher-order features were excluded to reduce the risk of overfitting given our sample size and to enhance the interpretability of the model. LASSO regression was used for feature selection. See [Supplementary 2 Section 2](#), for the feature extraction and selection process.

Spearman correlation analyses were performed after LASSO selection to assess associations between selected radiomic features and scanner parameters (slice thickness, tube voltage, scanner model, and scanning site), as well as with imaging findings (pancreas volume, calcifications, ductal dilations, and pseudocysts). Correlations with scanner parameters were examined to evaluate and mitigate potential scanner-related biases, whereas correlations with imaging findings were assessed to enhance the interpretability of the radiomic features. A Spearman correlation coefficient greater than 0.3 was considered indicative of a moderate association.

Training of radiomics classification models Random Forest, logistic regression, Gradient boost, XGBoost and support vector machine were tested for the best radiomics classifications model. The models were built for four tasks: CP classification (CP vs. healthy), EPI classification (yes/no), diabetes classification (yes/no), and pain classification (yes/no).

2.6. Model performance evaluation

Model performance metrics assessment included the area under the receiver operating characteristics curve (AUC), precision recall curve (PR-AUC) accuracy, precision, recall, specificity and F1 score. Ninety-five percent confidence intervals (CI) were estimated using bootstrap resampling with 2,000 repetitions. To enhance interpretability and clinical understanding of model characteristics, Shapley Additive Explanations (SHAP) were used with beeswarm summary plots ranking the overall importance of features in each model [13].

Classification probability scores The radiomics AI model provided a continuous probability score between 0 and 1 that quantifies how likely a CT scan belongs to the class of patients with CP or with a specific complication (EPI, diabetes or pain). To convert the probability scores into a binary clinical label, we selected the diagnostic cut-off value with maximized Youden's J statistic for the validation data.

EPI severity assessment: To explore whether the probability scores from the EPI model could be used to determine the severity of EPI, EPI probability scores were compared with the level of fecal elastase as a marker for exocrine function. Patients from the external test cohort were

stratified into a normal group ($\geq 200 \mu\text{g/g}$) and an EPI group ($< 200 \mu\text{g/g}$) and differences in probability distributions were assessed with the Wilcoxon rank-sum test, while the relationship between elastase values and EPI probability scores was quantified with Spearman's rank correlation.

Association with disease severity: To explore whether the EPI and diabetes probability scores of the AI model correlated with overall functional CP disease severity, we stratified patients from the external test cohort into three groups using endocrine and exocrine functional status as surrogate for cumulative CP severity: (group 0) preserved endocrine and exocrine functions, (group 1) failure of either endocrine or exocrine function, and (group 2) combined endocrine and exocrine failure. The probability score distributions were compared across these three groups using the Kruskal–Wallis test; when the test was significant, pairwise Wilcoxon rank-sum tests were performed with Holm correction applied to adjust for multiple comparisons. Results are reported as medians with inter-quartile ranges (IQRs) for each severity group.

3. Results

3.1. Study population

Demographical and clinical characteristics of the training- and test datasets are presented in [Table 1](#). In total, 201 CP patients and 148 pancreas-healthy controls from the Aalborg cohort were included for the training dataset; among the CP patients, 92% had definite CP and 8% had probable CP according to the M–ANNHEIM criteria. One patient was excluded from the training of the EPI and diabetes models, and nine were excluded from the pain model training due to missing clinical data. The test cohort ($n = 109$) consisted of 41 healthy controls from Aalborg and an external cohort of 68 CP patients from Bergen, among whom 79% had definite CP and 21% had probable CP.

3.2. Radiomics features

LASSO feature selection identified a total of 55 unique features to classify CP, EPI, diabetes and pain ([Table 2](#)). Twenty-one features correlated with pancreas volume, 18 with calcifications, and nine showed moderate correlations with ductal dilations, while none were associated with pseudocysts. Correlation analysis confirmed no scanner-related biases of the selected radiomics features. Both correlation analyses are presented in [Supplementary 2 Section 3](#),

3.3. Selection of AI model for training

A comprehensive comparison and analysis of the performance metrics for all classification models are provided in [Supplementary 2 Section 4](#). When considering overall performance metrics, XGBoost models based on 3D LASSO selected features demonstrated the highest overall efficacy; consequently, all results presented in this study are based on this model.

3.4. Classification performance of trained AI models

Full performance metrics are presented in [Table 3](#). The CP model achieved strong performance in the test dataset with an AUC of 0.97 (95% CI: 0.94–0.99). Complication-specific models demonstrated more variable performance, the EPI model showed moderate to strong performance with an AUC of 0.81 (95% CI: 0.66–0.88), while the diabetes and pain models had lower performance with an AUC of 0.63 (95% CI: 0.47–0.77) and 0.59 (95% CI: 0.37–0.67) respectively. Feature importance rankings for each model are presented in the SHAP summary plots ([Fig. 2](#)). The Youden J thresholds for the respective models can be seen in [Supplementary 2 Section 5](#).

Table 1

Clinical patient characteristics in the training cohorts and the test cohorts.

	Training cohorts		Test cohort	
	Aalborg CP (n = 201)	Aalborg healthy controls (n = 148)	Bergen CP (n = 68)	Aalborg healthy controls(n = 41)
Age, mean $\pm$ SD (range)	58.8 $\pm$ 11.8 (20–86)	54.0 $\pm$ 9.4 (20–81)	57.0 $\pm$ 13.3 (21–82)	53.1 $\pm$ 7.75 (40–68)
Sex				
Male, n	148 (71%)	103 (69%)	44 (55%)	27 (66%)
Female, n	61 (29%)	46 (31%)	36 (45%)	14 (34%)
Clinical characteristics				
Duration (yrs.) of CP, mean $\pm$ SD (range)	5.1 $\pm$ 6.8 (0–33)	–	2.7 $\pm$ 5.10 (0–27)	–
M–ANNHEIM diagnostic category, n (%)	Definite CP: 185/201 (92%) Probable CP: 16/201 (8%)	–	Definite CP: 54/68 (79%) Probable CP: 14/68 (21%)	–
EPI, n (%)	158/200 (79%) *	–	38/68 (56%)	–
Diabetes, n (%)	76/200 (38%) *	–	23/68 (34%)	–
Abdominal pain, n (%)	116/192 (60%) α	–	38/68 (56%)	–
CP etiology according to M–ANNHEIM criteria				
Alcohol, n (%)	112/201 (56%)	–	32/68 (47%)	–
Smoking, n (%)	155/201 (77%)	–	50/68 (74%)	–
Nutritional factors, n (%)	1/201 (0%)	–	3/68 (4%)	–
Efferent duct factors, n (%)	16/201 (8%)	–	22/68 (32%)	–
Immunological, n (%)	2/201 (1%)	–	3/68 (4%)	–
Misc./other, n (%)	10/201 (5%)	–	6/68 (8%)	–
Hereditary factors, n (%)	42/201 (21%)	–	7/68 (10%)	–
Unknown, n (%)	4/201 (2%)	–	9/68 (13%)	–
CT findings				
Calcifications, n (%)	142/201 (71%)	0 (0%)	27/68 (40%)	0 (0%)
Ductal dilations, n (%)	97/201 (48%)	0 (0%)	4/68 (6%)	0 (0%)
Pseudocysts, n (%)	48/201 (21%)	0 (0%)	10/68 (15%)	0 (0%)
Pancreas volume in ml, mean $\pm$ SD (range)	55.5 $\pm$ 33.8 (1.3 – 206.4)	65.9 $\pm$ 23.2 (17.9 – 121.6)	62.9 $\pm$ 21.1 (14.5 – 96.9)	73.1 $\pm$ 19.9 (29.7 – 115.0)

4. Association of AI model probability scores to fecal elastase and functional disease severity

In the test cohort from Bergen, two patients had missing fecal elastase values and were excluded. Among the remaining patients, 38 (58%) had EPI, while 28 (42%) had normal exocrine function. Patients with fecal elastase levels < 200 μ g/g (n = 38) had significantly higher EPI

model probability scores (median = 0.91, IQR = 0.25) compared to those with normal fecal elastase levels (median = 0.56, IQR = 0.39) (p = 0.001). Furthermore, EPI model probability scores were correlated with fecal elastase levels (Spearman's $r = -0.58$, $p < 0.001$), indicating that higher radiomics-based EPI probability scores were associated with lower fecal elastase values (Fig. 3).

The probability scores from both the EPI and diabetes models varied

Table 2

Radiomic features selected by the AI models for predicting CP, EPI, diabetes and pain.

Models	Features
CP model	- Shape: Sphericity, SurfaceVolumeRatio, SurfaceArea, LeastAxisLength, Elongation, Maximum2DDiameterSlice - First-order: Skewness, Uniformity, 10Percentile, Maximum, Median, Minimum - Texture: - GLCM: Idn - GLRLM: LongRunEmphasis- GLSZM: SizeZoneNonUniformityNormalized, LowGrayLevelZoneEmphasis
EPI model	- Shape: Maximum3DDiameter, MajorAxisLength, LeastAxisLength, Maximum2DDiameterSlice - First-order: Median, Kurtosis, InterquartileRangeMedian, Kurtosis, Uniformity, 10Percentile - Texture: - GLCM: MCC, Idn, Correlation - GLRLM: RunLengthNonUniformity - GLSZM: GrayLevelNonUniformityNormalized, LargeAreaLowGrayLevelEmphasis, LowGrayLevelZoneEmphasis, GrayLevelNonUniformity - GLDM: DependenceEntropy, DependenceNonUniformityNormalized- NGTDM: Busyness
Diabetes model	- Shape: Maximum2DDiameterColumn, Sphericity, Flatness, MinorAxisLength - First-order: Median, Kurtosis, InterquartileRange - Texture: - GLCM: ClusterProminence, InverseVariance - GLRLM: LongRunEmphasis, LowGrayLevelRunEmphasis - GLSZM: GrayLevelNonUniformity, LargeAreaHighGrayLevelEmphasis, SizeZoneNonUniformity, LargeAreaLowGrayLevelEmphasis, ZoneEntropy - GLDM: DependenceEntropy- NGTDM: Contrast, Strength
Pain model	- Shape: MeshVolume, Elongation, MinorAxisLength, Maximum2DDiameterRow - First-order: Median, Skewness, 10Percentile, Kurtosis, Uniformity - Texture: - GLCM: InverseVariance, Correlation, Autocorrelation, Idn - GLRLM: LongRunHighGrayLevelEmphasis, RunLengthNonUniformity - GLSZM: LowGrayLevelZoneEmphasis - GLDM: DependenceNonUniformityNormalized, SmallDependenceLowGrayLevelEmphasis- NGTDM: Coarseness, Contrast

Table 3
Performance metrics of the XGBoost AI models for predicting CP, EPI, Diabetes and Pain.

Model	Training dataset performance				Test dataset performance			
	CP	EPI	Diabetes	Pain	CP	EPI	Diabetes	Pain
AUC (95% CI)	0.97 (0.96–0.99)	0.76 (0.68–0.83)	0.73 (0.68–0.82)	0.57 (0.46–0.63)	0.97 (0.94–0.99)	0.81 (0.76–0.88)	0.63 (0.47–0.77)	0.59 (0.37–0.67)
PR-AUC (95% CI)	0.98 (0.97–0.99)	0.92 (0.88–0.96)	0.67 (0.55–0.76)	0.64 (0.55–0.74)	0.98 (0.96–1.0)	0.85 (0.67–0.92)	0.48 (0.37–0.73)	0.60 (0.43–0.74)
Accuracy (95% CI)	0.93 (0.91–0.96)	0.69 (0.62–0.75)	0.65 (0.60–0.72)	0.56 (0.49–0.64)	0.88 (0.82–0.94)	0.71 (0.59–0.79)	0.69 (0.55–0.71)	0.59 (0.38–0.63)
Balanced accuracy (95% CI)	0.93 (0.90–0.96)	0.73 (0.65–0.80)	0.69 (0.64–0.76)	0.58 (0.48–0.64)	0.89 (0.84–0.94)	0.70 (0.59–0.80)	0.55 (0.42–0.67)	0.49 (0.37–0.61)
Precision (95% CI)	0.95 (0.92–0.98)	0.93 (0.88–0.97)	0.52 (0.45–0.62)	0.66 (0.57–0.76)	0.97 (0.91–1)	0.77 (0.62–0.91)	0.39 (0.21–0.58)	0.55 (0.4–0.69)
Recall (Sensitivity) (95% CI)	0.93 (0.89–0.96)	0.65 (0.57–0.73)	0.87 (0.79–0.94)	0.57 (0.48–0.66)	0.84 (0.75–0.92)	0.63 (0.47–0.78)	0.48 (0.27–0.68)	0.61 (0.45–0.76)
Specificity (95% CI)	0.94 (0.90–0.97)	0.81 (0.68–0.92)	0.52 (0.44–0.62)	0.55 (0.44–0.67)	0.95 (0.86–1)	0.78 (0.61–0.91)	0.69 (0.48–0.76)	0.37 (0.2–0.54)
F1 score (95% CI)	0.93 (0.91–0.96)	0.77 (0.70–0.82)	0.65 (0.58–0.73)	0.58 (0.53–0.69)	0.90 (0.84–0.95)	0.70 (0.56–0.81)	0.43 (0.25–0.59)	0.58 (0.44–0.70)

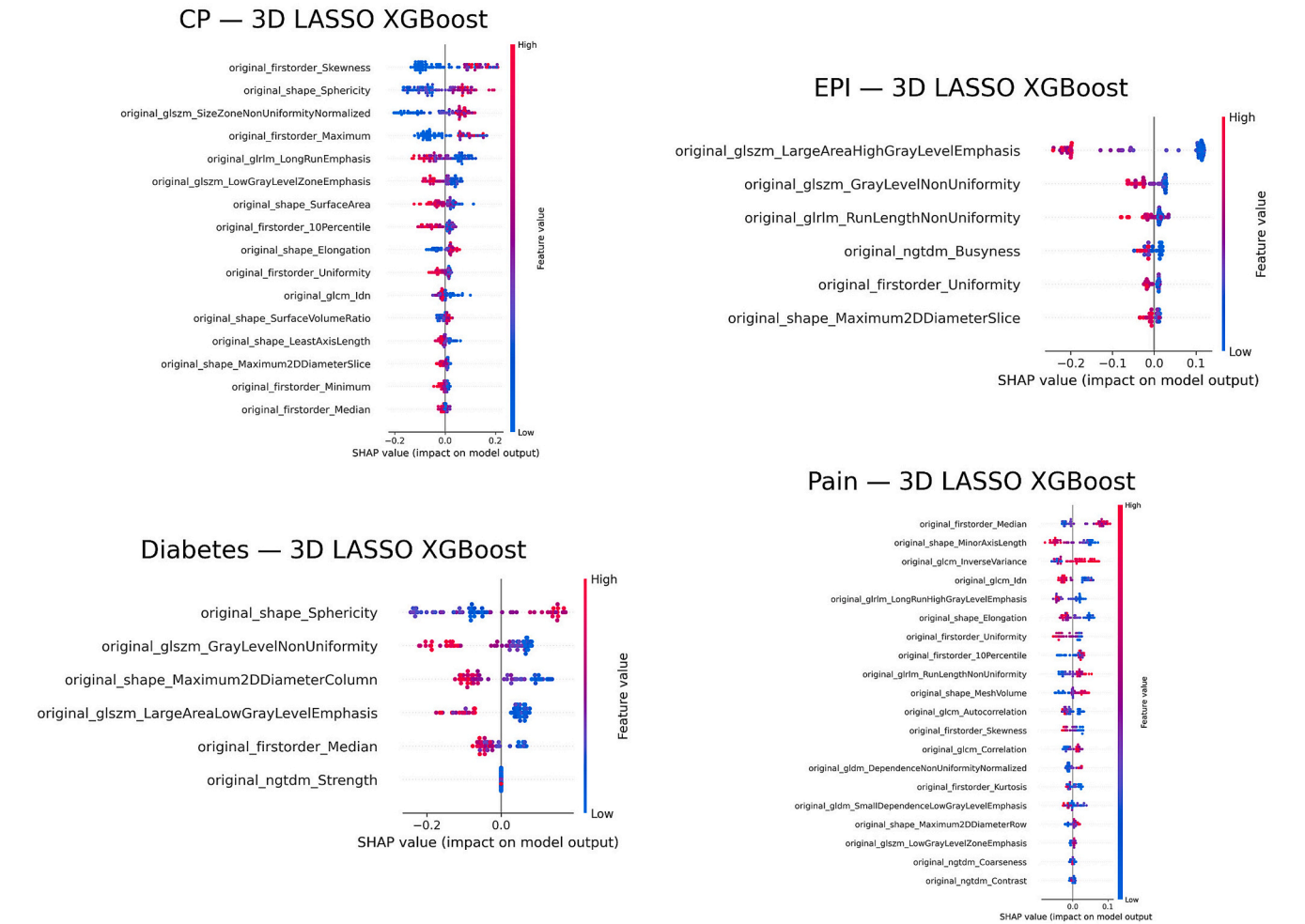


Fig. 2. SHAP (SHapley Additive exPlanations) beeswarm summary plots illustrating feature importance across the four AI models. Features are ranked by their overall impact, with the most influential displayed at the top.

significantly across the CP functional severity groups (Fig. 3). For the EPI model, a stepwise increase in EPI probability scores was seen with greater functional disease severity: median probability score was 0.52 (IQR = 0.35) in group 0, to 0.82 (IQR = 0.39) in group 1, and 0.95 (IQR = 0.09) in group 2. The difference among the groups were statistically significant ($p < 0.001$), and pairwise comparisons showed significant differences between group 0 and 1 ($p = 0.008$), between group 0 and 2 ($p < 0.001$), and between group 1 and 2 ($p = 0.01$). The diabetes model showed significant variation in probability scores across CP severity groups ($p = 0.004$). Pairwise comparisons revealed a significant difference between Group 0 and Group 2 ($p < 0.001$), while no significant differences were observed between Group 1 and either Group 0 or Group 2 ($p > 0.05$). Median probability scores increased with disease severity,

rising from 0.10 (IQR = 0.15) in Group 0 to 0.14 (IQR = 0.37) in Group 1, and 0.51 (IQR = 0.49) in Group 2.

5. Discussion

This study applied automatic pancreas segmentation and CT based radiomics analysis using AI models to classify CP and its associated complications. The CP model showed the strongest performance, while models classifying CP-related complications demonstrated moderate performance for EPI and more weak performance for diabetes and pain. In terms of feature interpretability, many of the selected radiomics features were associated with pancreatic volume, calcifications and ductal dilations. EPI probability scores were correlated with clinical

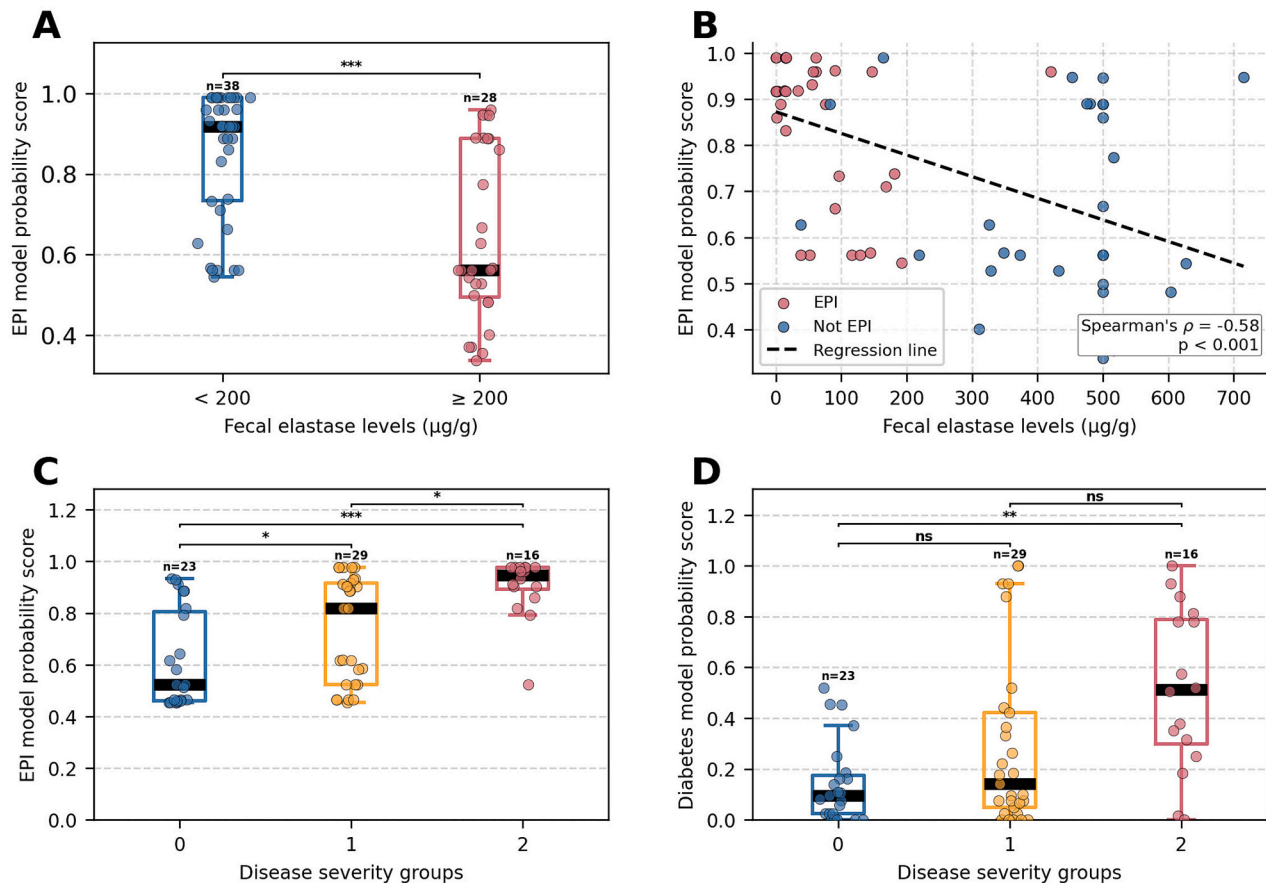


Fig. 3. (A) Box plot showing predicted probability scores from the exocrine pancreatic insufficiency (EPI) model between two groups defined by fecal elastase levels: < 200 µg/g (indicative of EPI) and ≥ 200 µg/g, with a significant difference observed. (B) Scatter plot of EPI model probability scores versus fecal elastase values, color-coded by elastase threshold. A significant negative correlation is observed (Spearman's $\rho = -0.58$, $p < 0.001$). (C) Box plot of EPI model probability scores stratified by disease severity based on their endocrine or exocrine insufficiency: group 0 = no functional impairment, group 1 = either endocrine or exocrine insufficiency, and group 2 = both endocrine and exocrine insufficiency. (D) Box plot of diabetes model probability scores stratified by the same disease severity groups as in (C). Distributions were compared using the Kruskal–Wallis test followed by pairwise Wilcoxon rank-sum tests. Sample sizes (n) are indicated above each group in all panels. Significance levels: ns = not significant, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

status such as fecal elastase levels and overall functional CP disease severity, highlighting the potential of radiomics models not only for classification but also for assessing disease severity. These findings may lay the groundwork for further discussions on the clinical relevance and limitations of AI radiomics-based tools in CP management.

5.1. Diagnostic potential of radiomics

The AI-based radiomics analysis identified features that largely reflected imaging markers of CP, including pancreatic volume and calcifications. Nonetheless, some features displayed weaker associations, implying that radiomics may offer complementary insights into pancreatic tissue characteristics beyond conventional imaging findings.

Radiomics-derived probability scores also correlated with clinical measures, including fecal elastase levels and functional disease severity. These associations suggest biological relevance and indicate that radiomics-derived features reflect disease burden on CT imaging. However, correlation alone does not establish clinical utility, and the present study does not demonstrate that these probability scores provide actionable information beyond existing clinical assessments. Nonetheless, these findings provide a rationale for further investigation of radiomics as a complementary imaging biomarker in prospective studies, that could support the development of a standardized, image-based severity scale that integrates structural features with functional outcome. Such an approach could enhance staging, enable personalized

treatment decisions, and provide reproducible biomarkers for monitoring progression and therapeutic response. Moreover, because radiomics likely detects early tissue heterogeneity and microstructural abnormalities, it holds promise for identifying CP in high-risk individuals before irreversible fibrosis or functional decline occurs [3,14].

For EPI model, the selected radiomics features related to pancreatic volume and morphology correlated strongly with fecal elastase, despite the model being trained only on whether the patients had EPI or not. The selected radiomics features also support the role of atrophy noted in previous studies [15]. Diagnosing EPI remains clinically challenging due to limitations in clinical tests such as fecal elastase measurements, secretin tests, and breath tests [10,16]. A supportive tool that applies radiomics to preexisting routine CT imaging could assist the clinicians in identifying patients at risk of having EPI, especially when the diagnosis is uncertain and clinical decision-making is challenging.

By contrast, the diabetes and pain models demonstrated poor performance. For diabetes, lack of distinction between type 1, type 2, and post-pancreatitis forms likely introduced heterogeneity that impaired predictive accuracy, and may also reflect that endocrine islet cell changes—representing only a small proportion of pancreatic tissue (1–2%)—is less radiologically visible on CT than the exocrine tissue changes (~90%) captured by the EPI model [17]. Pain prediction was even less reliable, with scattered low-impact features and no consistent imaging signature. This limitation reflects both the complexity of pain and the broad classification used, which did not differentiate chronic,

intermittent, or acute forms. Since pain is influenced by neuropathic or central mechanisms rather than pancreatic morphology alone, CT radiomics of the pancreas alone may be insufficient for accurate pain classification [18–20]. Future approaches may therefore benefit from multimodal strategies combining radiomics with MRI features related to neural or inflammatory changes and/or standardized clinical pain assessments to better capture the complex pathophysiology of pain in CP.

5.2. Comparison to previous studies

To date, not many studies have evaluated CT-based radiomics AI models specifically in CP. Frøkjær et al. investigated MR imaging radiomics in CP patients, reporting high accuracy rates (0.98 for CP, 0.83 for diabetes, and 0.82 for EPI) [21]. However, their reliance on a single 10-fold cross-validation raises concerns about overfitting and they had no external test dataset. In contrast, our study employed separate training, validation, and external testing cohorts, yielding more realistic performance estimates. The lower diabetes model performance compared with the MRI-based study by Frøkjær et al. could be multifactorial. MRI may better capture subtle parenchymal and ductal changes related to endocrine dysfunction, whereas CT predominantly reflects macroscopic structural alterations. In addition, cohort heterogeneity and the lack of diabetes subtype stratification in the present study may have reduced radiological signal associated with endocrine impairment.

Another study applied CT-based radiomics to differentiate among patients with abdominal pain, acute pancreatitis, and chronic pancreatitis, achieving an overall accuracy of 0.82 [22]. Nevertheless, the study was constrained by a small sample size ($n = 56$), employed leave-one-out cross-validation, and lacked an independent test set. Despite these limitations, the study identified gray-level co-occurrence matrix features as particularly informative of distinguishing the three groups of patients [22].

Despite the growing interest in radiomics for oncologic applications, studies focusing specifically on chronic inflammatory pancreatic diseases remain scarce. A greater emphasis on radiomics research in CP has the potential to enhance diagnostic accuracy from clinical CT scans and inform clinical decision-making in this underrepresented area.

5.3. Strengths and limitations

A major strength of this study is the multicenter inclusion of a diverse range of CT scanner models and acquisition parameters, which enhances the generalizability of our findings. Additionally, validation using an independent test dataset with external CP patients supports the reproducibility of our model. However, several limitations should be noted. As a retrospective cross-sectional study, our data may not fully reflect real-world clinical complexity. We also lacked follow-up assessments to confirm the accuracy of clinical disease labels. Furthermore, differences in clinical practice, even though the same M–AANHEIM criteria were used, may have introduced inconsistencies in our dataset or contributed to misclassification of CP severity. In particular, the accuracy of our AI models may have been affected by imprecise or incomplete data on pain and EPI. However, these limitations also reflect the complexities and imperfections of real-world clinical data. Finally, a key limitation of this study is the absence of direct comparison with radiologist interpretation of CT scans or established imaging-based diagnostic classifications. As a result, the present work cannot determine whether radiomics provides incremental diagnostic value beyond current clinical imaging assessment. Accordingly, this study should be regarded as a technical proof-of-concept demonstrating the feasibility of CT-based radiomics in chronic pancreatitis.

5.4. Clinical implications and future directions

Prospective studies are warranted to validate the clinical utility of radiomics for early CP detection and dynamic monitoring of disease progression. Future research should also investigate whether radiomics-derived features can monitor and potentially predict treatment responses, such as outcomes following enzyme replacement, endoscopic interventions or future potential anti-fibrotic treatments, and improve personalized management by integrating imaging with clinical, serological, and genetic biomarkers. Ultimately, creating prospective, multi-center databases with standardized imaging and clinical data, including longitudinal follow-up, will be critical for establishing AI-driven tools that support early diagnosis, monitoring and personalized management of CP.

6. Conclusion

Our study demonstrated that a CT radiomics-based AI model can effectively classify patients based on CP diagnosis. Notably, radiomics-based probability scores correlated significantly with both functional CP disease severity and fecal elastase values, suggesting that these scores could serve as a quantitative measure of disease severity. The specific AI models for complications of EPI, diabetes, and pain yielded variable results, highlighting both the promise and the challenges of applying radiomics to complex CP manifestations. Despite limitations, our findings support the feasibility of radiomics as imaging biomarkers for CP. With further methodological refinements and prospective validation, radiomics-based AI models incorporating automatic pancreatic segmentation have the potential to transform personalized CP management through earlier diagnosis, improved monitoring, enhanced risk stratification, and tailored therapeutic interventions.

CRediT authorship contribution statement

Surenth Nalliah: Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Søren Nicolai Frederiksen Hostrup:** Writing – review & editing, Software, Methodology. **Esben Bolvig Mark:** Writing – review & editing, Software, Methodology. **Marjolein Henrieke Liedenaum:** Writing – review & editing, Validation, Data curation. **Erlend Hodneland:** Writing – review & editing, Validation, Software. **Ingfrid Helene Salvesen Haldorsen:** . **Trond Engjom:** Writing – review & editing, Validation, Supervision, Conceptualization. **Asbjørn Mohr Drewes:** Writing – review & editing, Supervision, Resources. **Søren Schou Olelsen:** Writing – review & editing, Supervision, Resources. **Jens Brøndum Frøkjær:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ejrad.2026.112711>.

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